

Theory of Computation

Exercise 4:

(Closure properties of Regular Language and Regular Expression)

1. Prove that the language $\{a^m b : m \geq 1 \text{ and } m \neq 100\}$ is regular.

$$aa^+b \cap \overline{a^{100}b}$$

2. Find regular expression for the following language

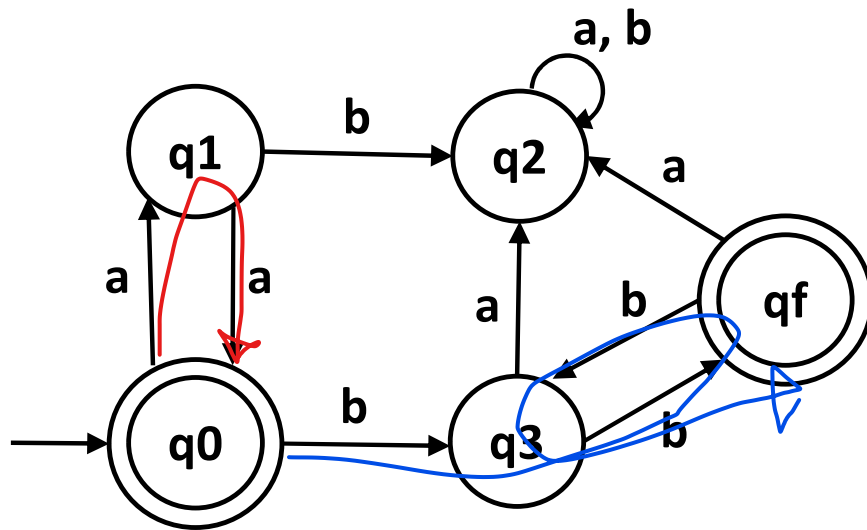
$$L = \{ w \in \{a, b\}^* : w \text{ does not end with } \underline{ab} \}$$

~~$(a+b)^* \wedge (a+b)^* \overline{ab}$~~

$(a+b)^* \cdot (aa+ba+bb)^+ \cup a+b$

*3. Find regular expression for the following DFA.

(Homework 3)



$$(aa)^* + bb(bb)^*$$